

Network Reconnaissance Using Nmap

Project Report
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Abstract

This project demonstrates network reconnaissance and vulnerability assessment using Nmap, Metasploit Framework, and Nessus Essentials within a controlled lab environment.

Objectives

- Understand reconnaissance techniques
- Perform service and OS detection
- Integrate Metasploit with Nmap
- Perform vulnerability assessment using Nessus

Tools Used

Kali Linux
Nmap
Metasploit Framework
Nessus Essentials
Apache2
OpenSSH
Netcat
VMware

Lab Setup

Configured Services:

- Port 22 - SSH
- Port 80 - Apache HTTP Server
- Port 8080 - Python HTTP Server
- Port 4444 - Netcat Listener
- Port 9999 - Netcat Listener

Nmap Commands

```
nmap 127.0.0.1  
sudo nmap -sV 127.0.0.1  
sudo nmap -A 127.0.0.1  
sudo nmap -O 127.0.0.1
```

Metasploit Integration

Commands Used:

- msfconsole
- db_status
- db_nmap 127.0.0.1
- hosts
- services

Nessus Vulnerability Assessment

Nessus Essentials successfully identified multiple informational and service-related findings including Apache HTTP issues, SSH findings, SSL/TLS observations, and service enumeration data.

Conclusion

The project successfully demonstrated practical network reconnaissance and vulnerability assessment techniques in a controlled cybersecurity environment.

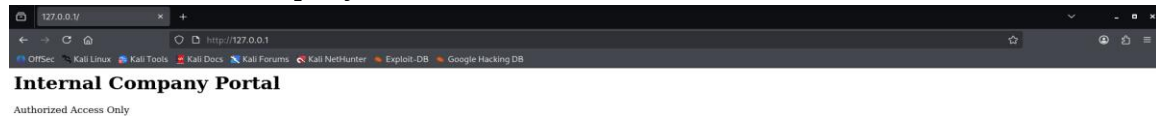
Screenshots Section

Apache Web Server Running

```
[sudo] password for prkhar:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
   Active: active (running) since Wed 2026-05-27 02:25:53 PDT; 2h 58min ago
 Invocation: 4e8740a7ad514e8a8440f2026b69462c
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 4238 (apache2)
   Status: "Total requests: 1506; Idle/Busy workers 100/0;Requests/sec: 0.14;>
    Tasks: 9 (limit: 2073)
  Memory: 7.8M (peak: 32M, swap: 12.2M, swap peak: 13.5M)
     CPU: 1.261s
    CGroup: /system.slice/apache2.service
           └─ 4238 /usr/sbin/apache2 -k start -DFOREGROUND
              4254 /usr/sbin/apache2 -k start -DFOREGROUND
              4255 /usr/sbin/apache2 -k start -DFOREGROUND
              4256 /usr/sbin/apache2 -k start -DFOREGROUND
              4258 /usr/sbin/apache2 -k start -DFOREGROUND
              4259 /usr/sbin/apache2 -k start -DFOREGROUND
             14520 /usr/sbin/apache2 -k start -DFOREGROUND
             22411 /usr/sbin/apache2 -k start -DFOREGROUND
             88204 /usr/sbin/apache2 -k start -DFOREGROUND

May 27 02:25:53 kali systemd[1]: Starting apache2.service - The Apache HTTP Ser>
May 27 02:25:53 kali apachectl[4238]: AH00558: apache2: Could not reliably dete>
```

Custom Internal Company Portal



Nmap Basic Scan

```
(prkhar@kali)-[~]
$ sudo nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 02:51 -0700
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000010s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
3306/tcp  open  mysql
4444/tcp  open  krb524
8080/tcp  open  http-proxy
9999/tcp  open  abyss

Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
```

Nmap Service Detection Scan

```
(prkhar@kali)~$ sudo nmap -T4 -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 02:48 -0700
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000010s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 10.3p1 Debian 1 (protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.67 ((Debian))
3306/tcp  open  mysql?
4444/tcp  open  krb524?
8080/tcp  open  http     SimpleHTTPServer 0.6 (Python 3.13.9)
9999/tcp  open  abyss?
1 service unrecognized despite returning data. If you know the service/version, please submit the
following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port3306-TCP:V=7.99%I=7%D=5/27%Time=6A168E11%P=x86_64-pc-linux-gnu%r(NU
SF:LL,64,"`\0\0\0\n11\8\6-MariaDB-6\x20from\x20Debian\0[\0\0\0009l7l:ys
SF:Y\0\0\xfe\xff-\x02\0\xff\x81\x15\0\0\0\0\0=\0\0\0Ax\[\0?feIzMQ09\0mysql_
SF:native_password\0")%r(GenericLines,9C,"`\0\0\0\n11\8\6-MariaDB-6\x20f
SF:rom\x20Debian\0[\0\0\0009l7l:ysY\0\0\xfe\xff-\x02\0\xff\x81\x15\0\0\0\0\
SF:0\0-\0\0\0Ax\[\0?feIzMQ09\0mysql_native_password\x004\0\0\01\xffj\x04#H
SF:Y000Proxy\x20header\x20is\x20not\x20accepted\x20from\x20127\0\0\0.1")%
SF:r(LDAPBindReq,64,"`\0\0\0\n11\8\6-MariaDB-6\x20from\x20Debian\0\0\0\0\
SF:0w6E8`Ym8\0\0\xfe\xff-\x02\0\xff\x81\x15\0\0\0\0\0=\0\0\0nB>Q\`aL;!0nZ\
SF:0mysql_native_password\0")%r(afp,64,"`\0\0\0\n11\8\6-MariaDB-6\x20fro
SF:m\x20Debian\0v\0\0\0je;8G:\.w\0\0\xfe\xff-\x02\0\xff\x81\x15\0\0\0\0\0=
SF:\0\0\0x000(s/\0.7h)\1[Uz\0mysql_native_password\0");
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.29 seconds
```

Nmap Aggressive Scan

```
(prkhar@kali)~$ sudo nmap -T4 -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 02:55 -0700
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000010s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 10.3p1 Debian 1 (protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.67 ((Debian))
|_http-title: Site doesn't have a title (text/html).
|_http-server-header: Apache/2.4.67 (Debian)
3306/tcp  open  mysql?
|_ssl-cert: Subject: commonName=MariaDB Server
|_not valid before: 2026-05-27 09:26:13
|_not valid after: 2036-05-24 09:26:13
|_ssl-date: TLS randomness does not represent time
Fingerprint-strings:
|_GenericLines:
|_11.8.6-MariaDB-6 from Debian
|_m0l2l0
|_0Q01C6/8cn
|_mysql_native_password
|_MySQLProxy header is not accepted from 127.0.0.1
|_LDAPBindReq
|_11.8.6-MariaDB-6 from Debian
|_Aafn12
|_MySQL-gm5j
|_mysql_native_password
|_Null:
|_11.8.6-MariaDB-6 from Debian
|_m0l2l0
|_0Q01C6/8cn
|_mysql_native_password
|_afp:
|_11.8.6-MariaDB-6 from Debian
|_vAcshV0
|_wX00w7j3tc
|_mysql_native_password
|_mysql-info:
|_Protocol: 10
|_Version: 11.8.6-MariaDB-6 from Debian
|_Thread ID: 151
|_Capabilities flags: 65534
|_Some Capabilities: SupportsAuth, SupportsProtocolOld, LongColumnFlag, DBCCClient, SwitchToSSLAfterHandshake, DontAllowDatabaseFalcColumn, SupportsCompression, SupportsTransactions, IgnoreSpaceBeforeParenthesis, InteractiveClient,
ConnectWithDatabase, FoundRows, SupportsLoadDataLocal, IgnoreSigpipes, SupportsProtocolNew, SupportsMultipleStatements, SupportsAuthPlugins, SupportsMultipleResults
|_Status: Autocommit

|_Status: Autocommit
|_Salt: 8c2f24fc-c188-549e
|_Auth Plugin Name: mysql_native_password
4444/tcp  open  krb524?
8080/tcp  open  http     SimpleHTTPServer 0.6 (Python 3.13.9)
|_http-title: Directory listing for /
|_http-server-header: SimpleHTTP/0.6 Python/3.13.9
9999/tcp  open  abyss?
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port3306-TCP:V=7.99%I=7%D=5/27%Time=6A168E11%P=x86_64-pc-linux-gnu%r(NU
SF:LL,64,"`\0\0\n11\8\6-MariaDB-6\x20from\x20Debian\0[\0\0\0m0b120l2l0
SF:Y\0\0\xfe\xff-\x02\0\xff\x81\x15\0\0\0\0\0=\0\0\0Ax\[\0?feIzMQ09\0mysql_
SF:native_password\0")%r(GenericLines,9C,"`\0\0\0\n11\8\6-MariaDB-6\x20f
SF:rom\x20Debian\0[\0\0\0m0b120l2l0:ysY\0\0\xfe\xff-\x02\0\xff\x81\x15\0\0\0\0\
SF:0\0-\0\0\0Ax\[\0?feIzMQ09\0mysql_native_password\x004\0\0\01\xffj\x04#H
SF:Y000Proxy\x20header\x20is\x20not\x20accepted\x20from\x20127\0\0\0.1")%
SF:r(LDAPBindReq,64,"`\0\0\0\n11\8\6-MariaDB-6\x20from\x20Debian\0\0\0\0\
SF:0w6E8`Ym8\0\0\xfe\xff-\x02\0\xff\x81\x15\0\0\0\0\0=\0\0\0nB>Q\`aL;!0nZ\
SF:0mysql_native_password\0")%r(afp,64,"`\0\0\0\n11\8\6-MariaDB-6\x20fro
SF:m\x20Debian\0v\0\0\0je;8G:\.w\0\0\xfe\xff-\x02\0\xff\x81\x15\0\0\0\0\0=
SF:\0\0\0x000(s/\0.7h)\1[Uz\0mysql_native_password\0");
Device type: general purpose
Running: Linux 5.x16.x
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.53 seconds
```

Metasploit db_nmap Results

```
msfconsole
Metasploit tip: Metasploit can be configured at startup, see msfconsole
--help to learn more

[... ASCII art ...]

= [ metasploit v6.4.99-dev ]
+ -- -- [ 2,572 exploits - 1,317 auxiliary - 1,683 payloads ]
+ -- -- [ 433 post - 49 encoders - 13 nops - 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > db_status
[*] Connected to msf. Connection type: postgresql.
msf > db_nmap 127.0.0.1
[*] Nmap: Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 05:29 -0700
[*] Nmap: Nmap scan report for localhost (127.0.0.1)
[*] Nmap: Host is up (0.0000010s latency).
[*] Nmap: Not shown: 995 closed tcp ports (reset)
[*] Nmap: PORT      STATE SERVICE
[*] Nmap: 21/tcp    open  ftp
[*] Nmap: 22/tcp    open  ssh
[*] Nmap: 80/tcp    open  http
[*] Nmap: 3306/tcp  open  mysql
[*] Nmap: 5432/tcp  open  postgresql
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
msf > hosts

Hosts
=====
Order: Time-Based

address    mac      name      os_name  os_flavor  os_sp  purpose  info  comments
-----
127.0.0.1  localhost Unknown  device

msf >
```

Nessus Host Scan Summary

My Basic Network Scan

Hosts: 1 | Vulnerabilities: 75% | History: 1

Host	Auth	Vulnerabilities
127.0.0.1	Pass	0

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVEs v3.0
- Scanner: Local Scanner
- Start: Today at 5:02 AM
- End: Today at 5:15 AM
- Elapsed: 13 minutes

Vulnerabilities

- 0 Critical
- 0 High
- 0 Medium
- 0 Low
- 0 Info

Nessus Vulnerability Findings

The screenshot displays the Tenable Nessus Essentials Plus interface. The left sidebar contains navigation options: FOLDERS (My Scans, All Scans, Trash), RESOURCES (Policies, Plugin Rules), and Tenable News. The main panel shows a table of 75 vulnerabilities. The table columns are: Filter, Search Vulnerabilities, CVSS, VPR, EPSS, Name, Family, and Count. The vulnerabilities listed include: Python Library Brotli <= 1.1.0 DoS (HIGH, 7.5, 3.6, 0.0004), Apache HTTP Server (Multiple Issues) (MEDIUM), SSH (Multiple Issues) (INFO), HTTP (Multiple Issues) (INFO), DMI (Multiple Issues) (INFO), SSH (Multiple Issues) (INFO), Nodejs Node.js (Multiple Issues) (INFO), SSH (Multiple Issues) (INFO), SSL (Multiple Issues) (INFO), TLS (Multiple Issues) (INFO), Remote listeners enumeration (Linux / AIX) (INFO), Netstat Portscanner (SSH) (INFO), and Service Detection (INFO). The right sidebar shows Scan Details for a Basic Network Scan, including Status (Completed), Severity Base (CVSS v3.0), Scanner (Local Scanner), Start (Today at 5:02 AM), End (Today at 5:15 AM), and Elapsed (13 minutes). Below this is a Vulnerabilities donut chart showing the distribution of severity levels: Critical (red), High (orange), Medium (yellow), Low (green), and Info (blue).

Filter	Search Vulnerabilities	CVSS	VPR	EPSS	Name	Family	Count
Sev		7.5	3.6	0.0004	Python Library Brotli <= 1.1.0 DoS	Misc.	1
Sev					Apache HTTP Server (Multiple Issues)	Web Servers	4
Sev					SSH (Multiple Issues)	General	9
Sev					HTTP (Multiple Issues)	Web Servers	5
Sev					DMI (Multiple Issues)	General	3
Sev					SSH (Multiple Issues)	Misc.	3
Sev					Nodejs Node.js (Multiple Issues)	Misc.	2
Sev					SSH (Multiple Issues)	Service detection	2
Sev					SSL (Multiple Issues)	General	2
Sev					TLS (Multiple Issues)	General	2
Sev					Remote listeners enumeration (Linux / AIX)	Service detection	6
Sev					Netstat Portscanner (SSH)	Port scanners	5
Sev					Service Detection	Service detection	5

References

- [Official Nmap Documentation](#)
- [Metasploit Framework Documentation](#)
- [Tenable Nessus Documentation](#)
- [Kali Linux Documentation](#)